

Canadian Bank Market Risk VaR Engine

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Executive Summary

The Canadian Bank Market Risk VaR Engine evaluates the market risk of an equal-weight portfolio of six major Canadian banks with an initial notional value of CAD \$1,000,000. The analysis combines Value at Risk (VaR), Expected Shortfall (ES), dynamic volatility modelling, out-of-sample backtesting, and historical and hypothetical stress testing.

Four one-day VaR methodologies are evaluated in the rolling backtest: Historical Simulation, Rolling Gaussian VaR, EWMA Gaussian VaR, and GARCH(1,1) Gaussian VaR. At the 99% confidence level, the highest-ranked model under the dashboard's descriptive validation criteria is **Historical Simulation**, which passes 1 of the three statistical tests considered. This ranking should not be interpreted as definitive model selection; model performance must also be evaluated in terms of stability, responsiveness, tail assumptions, and economic usefulness.

The latest 99% VaR estimate from this model is approximately **CAD \$21,337.60**. This represents a model-implied one-day loss threshold rather than a maximum possible portfolio loss.

Tail-risk analysis demonstrates the importance of examining losses beyond the VaR boundary. Full-sample 99% Historical Expected Shortfall is approximately **CAD \$45,335.28**, measuring the average historical loss conditional on entering approximately the worst 1% of the return distribution.

Stress testing produces substantially different information from probabilistic VaR. The largest loss across the stress scenarios considered is approximately **CAD \$392,163**, illustrating that severe scenario losses can materially exceed ordinary one-day VaR estimates.

The results also highlight an important concentration risk. Although the portfolio contains six securities, every constituent belongs to the Canadian banking sector. Security-level diversification therefore does not eliminate exposure to systemic banking, credit, interest-rate, or macroeconomic shocks affecting the sector as a whole.

Portfolio Definition

The baseline portfolio contains six major Canadian publicly traded banks. Each security initially receives an equal portfolio weight.

Table 1: Executive Market Risk Snapshot

Risk Measure	Value
Portfolio Value	CAD \$1,000,000
Latest EWMA Annualized Volatility	16.21%
Latest GARCH Annualized Volatility	17.09%
Latest 99% Historical VaR	CAD \$21,338
Latest 99% GARCH VaR	CAD \$19,103
Full-Sample 99% Historical ES	CAD \$45,335
Full-Sample 99% Gaussian ES	CAD \$28,211
Largest Stress Loss	CAD \$392,163

The initial portfolio value is CAD \$1,000,000. Risk will be measured over a 1-day horizon at the 95% and 99% confidence levels.

Data and Methodology

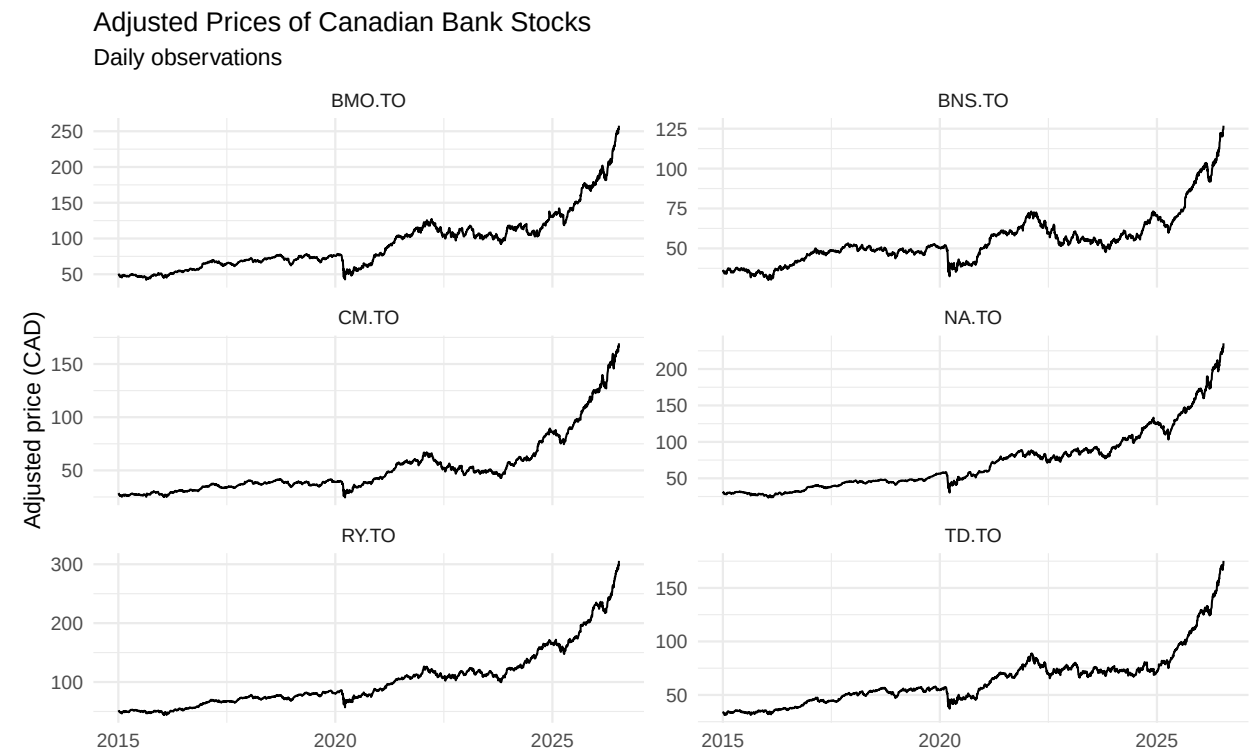
Daily stock-price data are obtained from Yahoo Finance using the `tidyquant` package. Adjusted closing prices are used so that the return series reflects relevant corporate actions, including stock splits and dividend distributions.

Data Coverage

Table 2: Historical Market-Data Coverage

Ticker	Bank	First Observation	Last Observation	Observations	Missing Adjusted Prices	Duplicate Dates
BMO.TO	Bank of Montreal	2015-01-02	2026-07-15	2,895	0	0
BNS.TO	Bank of Nova Scotia	2015-01-02	2026-07-15	2,895	0	0
CM.TO	Canadian Imperial Bank of Commerce	2015-01-02	2026-07-15	2,895	0	0
NA.TO	National Bank of Canada	2015-01-02	2026-07-15	2,895	0	0
RY.TO	Royal Bank of Canada	2015-01-02	2026-07-15	2,895	0	0
TD.TO	Toronto-Dominion Bank	2015-01-02	2026-07-15	2,895	0	0

Adjusted Stock Prices



Return Construction

For each bank, the daily simple return is calculated as

$$R_{i,t} = \frac{P_{i,t}}{P_{i,t-1}} - 1,$$

where $P_{i,t}$ is the adjusted closing price of asset i on date t .

The corresponding continuously compounded return is

$$r_{i,t} = \ln(1 + R_{i,t}).$$

The baseline portfolio is treated as a daily-rebalanced, constant-weight portfolio. Its simple return is therefore

$$R_{p,t} = \sum_{i=1}^N w_i R_{i,t}.$$

The portfolio log return is calculated from the aggregated simple return:

$$r_{p,t} = \ln(1 + R_{p,t}).$$

Return-Data Alignment

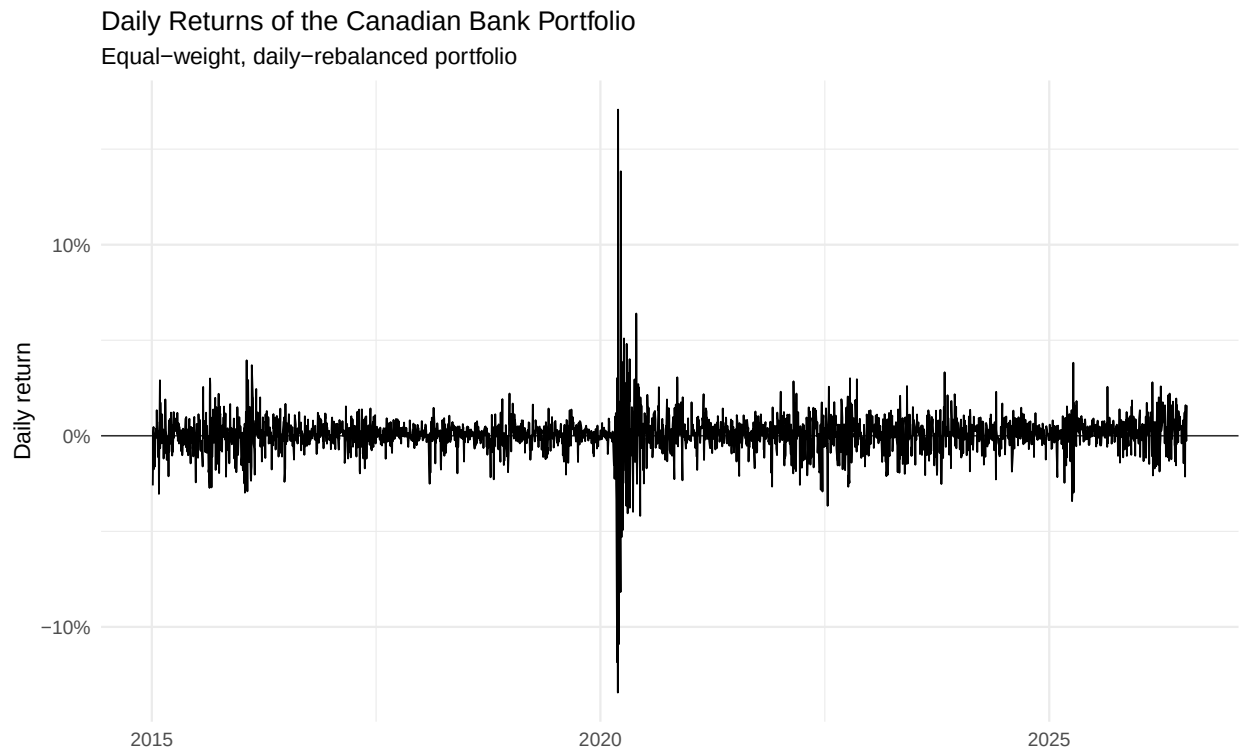
Only dates containing valid returns for all six portfolio assets are retained.

Table 3: Return-Panel Alignment Summary

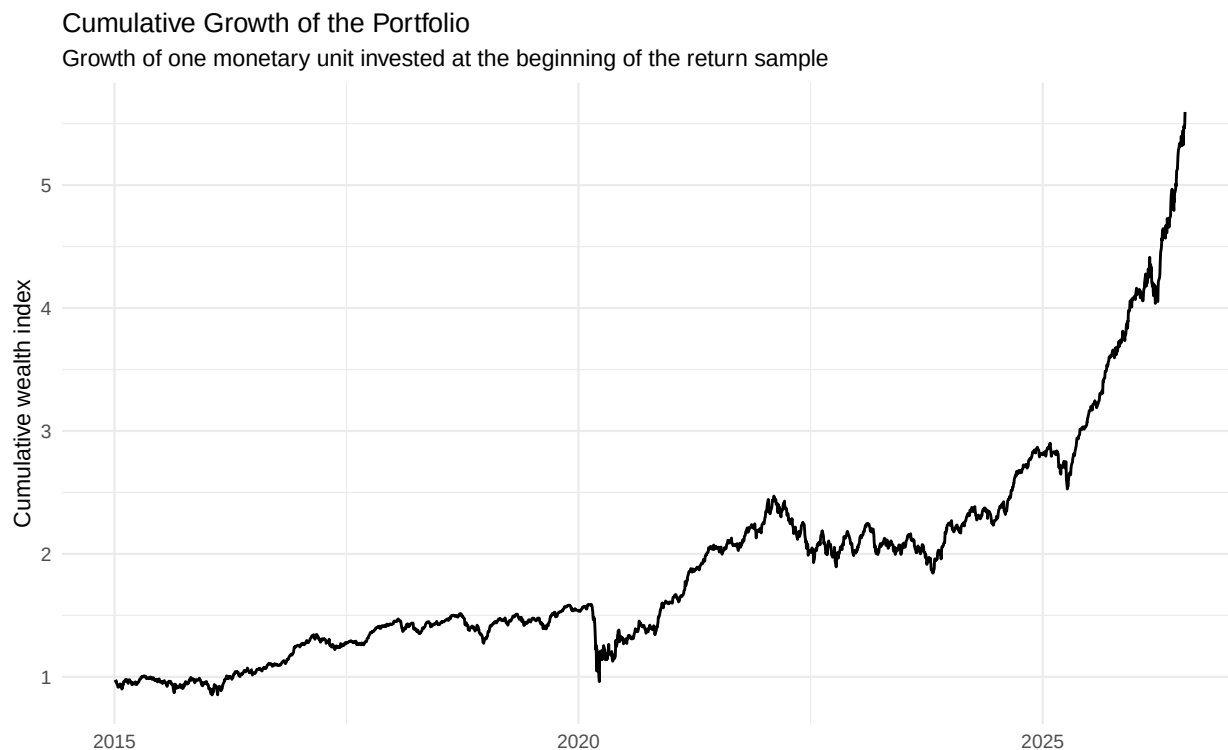
Number of Assets	Candidate Return Dates	Common Return Dates	Dates Removed
6	2894	2894	0

Exploratory Analysis

Daily Portfolio Returns



Cumulative Portfolio Performance



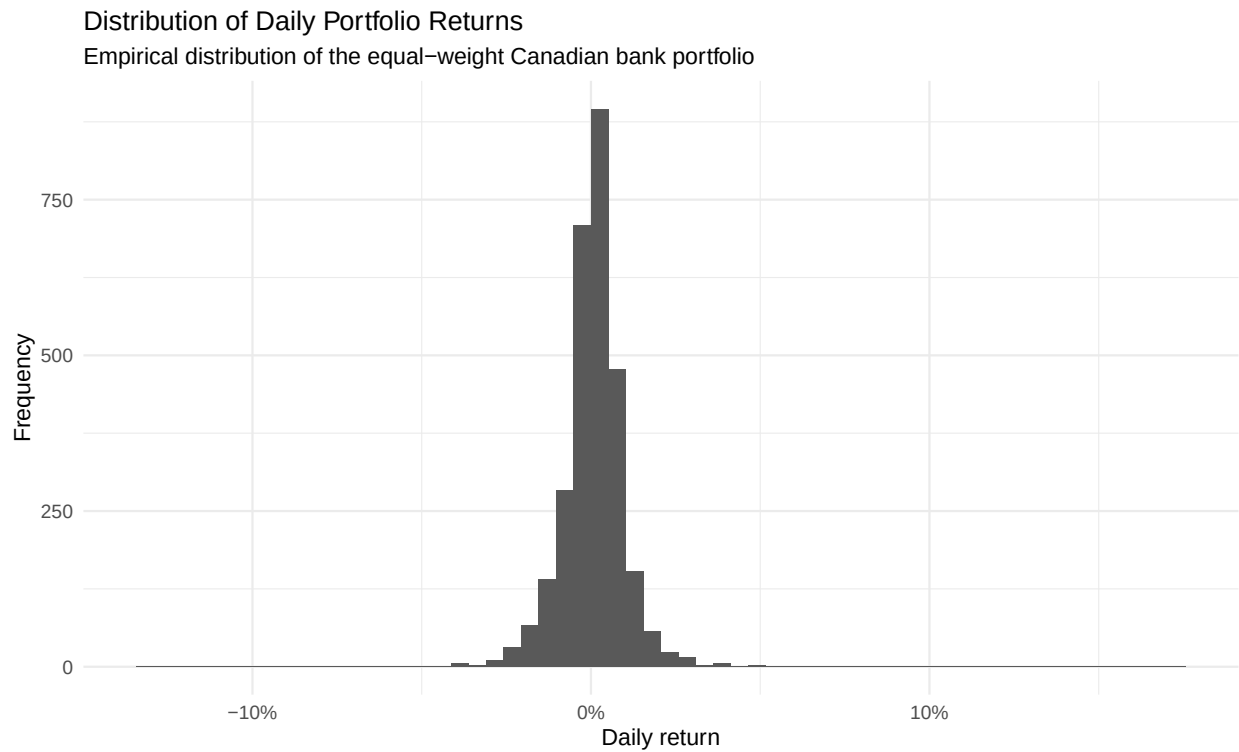
Descriptive Risk Statistics

Before estimating Value at Risk, the empirical distribution of portfolio returns is examined. Particular attention is given to volatility, skewness, kurtosis, extreme observations, and dependence among the constituent securities.

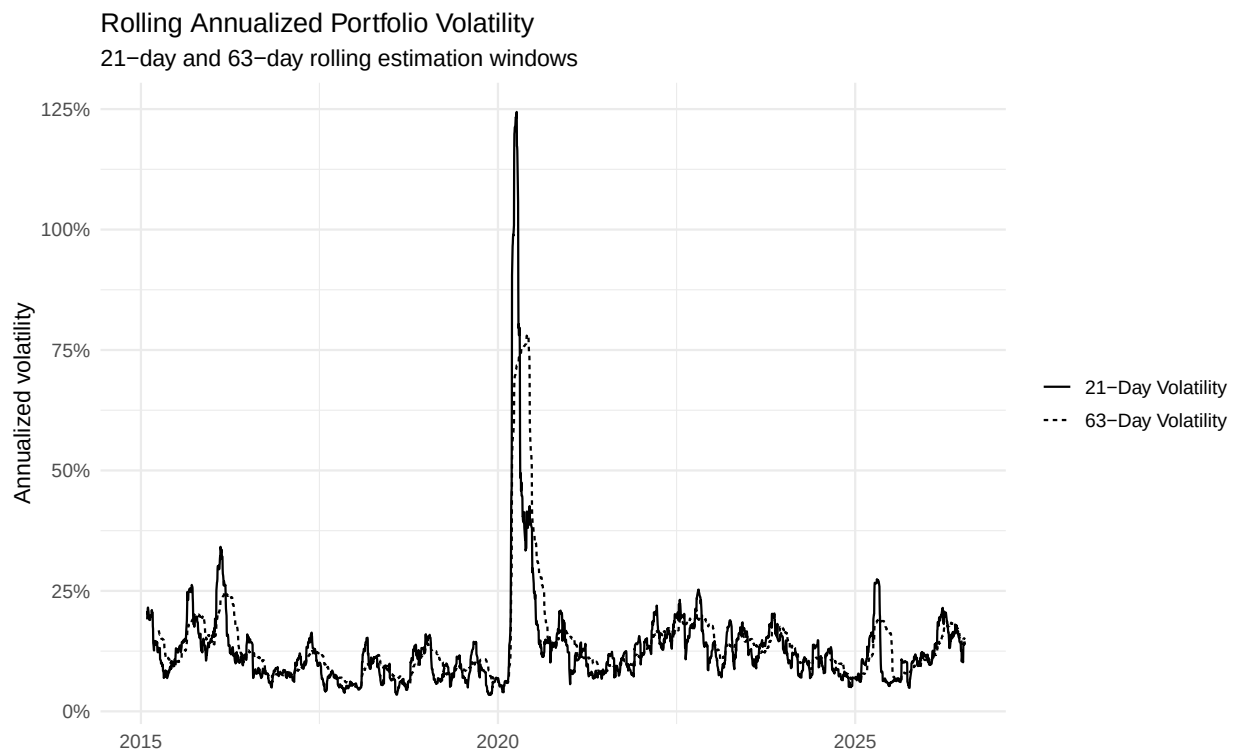
Table 4: Portfolio Descriptive Risk Statistics

Observations	Annualized Return	Annualized Volatility	Skewness	Excess Kurtosis	Worst Daily Return	Best Daily Return	Maximum Drawdown
2894	16.18%	17.19%	0.41	50.271	-13.44%	17.07%	-39.46%

Distribution of Portfolio Returns



Rolling Portfolio Volatility



Cross-Asset Correlations

Table 5: Daily Return Correlations Among Canadian Banks

ticker	RY.TO	TD.TO	BMO.TO	BNS.TO	CM.TO	NA.TO
RY.TO	1.000	0.758	0.776	0.766	0.771	0.734
TD.TO	0.758	1.000	0.751	0.756	0.708	0.693
BMO.TO	0.776	0.751	1.000	0.776	0.783	0.725
BNS.TO	0.766	0.756	0.776	1.000	0.778	0.714
CM.TO	0.771	0.708	0.783	0.778	1.000	0.704
NA.TO	0.734	0.693	0.725	0.714	0.704	1.000

Largest Historical Daily Losses

Table 6: Ten Worst Portfolio Trading Days

Date	Portfolio Return	Portfolio P&L
2020-03-12	-13.44%	-CAD \$134,355
2020-03-09	-11.84%	-CAD \$118,438
2020-03-16	-10.90%	-CAD \$109,023
2020-03-23	-8.18%	-CAD \$81,758
2020-03-18	-6.39%	-CAD \$63,851
2020-03-27	-5.31%	-CAD \$53,061
2020-04-01	-4.93%	-CAD \$49,328
2020-06-11	-4.19%	-CAD \$41,916
2020-04-21	-4.06%	-CAD \$40,564
2020-05-13	-3.98%	-CAD \$39,802

Value at Risk

Value at Risk (VaR) estimates the loss threshold associated with a specified confidence level and risk horizon. The analysis considers one-day VaR at both the 95% and 99% confidence levels.

Two approaches are compared: Historical Simulation and a Parametric Gaussian variance-covariance model.

Historical Simulation VaR

Historical Simulation makes no parametric distributional assumption. For confidence level c , VaR is estimated from the empirical lower tail of historical portfolio returns:

$$VaR_c = -Q_{1-c}(R_p).$$

Thus, 95% VaR corresponds to the negative of the empirical 5th percentile, while 99% VaR corresponds to the negative of the empirical 1st percentile.

Parametric Gaussian VaR

The parametric model assumes that daily portfolio returns follow a normal distribution:

$$R_p \sim N(\mu_p, \sigma_p^2).$$

Portfolio volatility is calculated using the asset covariance matrix:

$$\sigma_p = \sqrt{\mathbf{w}'\Sigma\mathbf{w}}.$$

The resulting VaR is

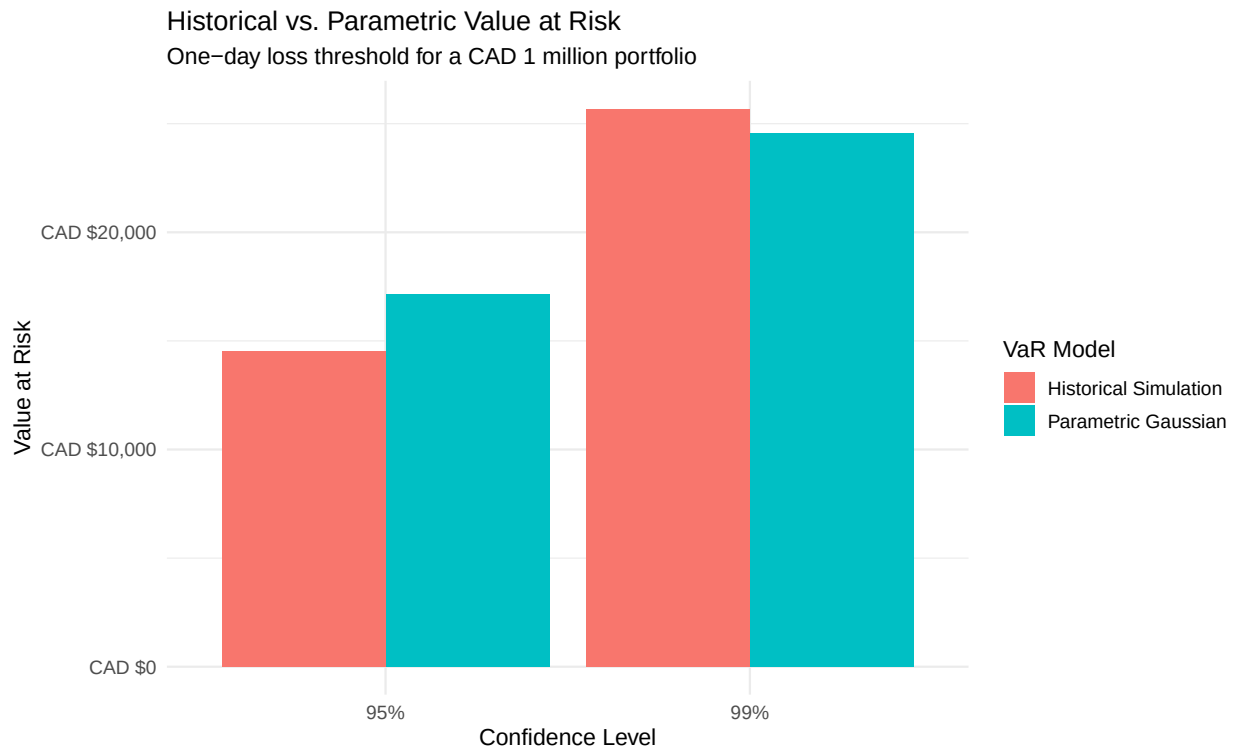
$$VaR_c = -[\mu_p + z_{1-c}\sigma_p].$$

VaR Estimates

Table 7: One-Day Value at Risk Estimates

Model	Confidence Level	VaR (%)	VaR (CAD)
Historical Simulation	95%	1.45%	CAD \$14,512.29
Parametric Gaussian	95%	1.72%	CAD \$17,160.48
Historical Simulation	99%	2.57%	CAD \$25,689.00
Parametric Gaussian	99%	2.45%	CAD \$24,541.25

Comparison of VaR Models



Expected Shortfall

Expected Shortfall (ES), also referred to as Conditional Value at Risk (CVaR), measures the average loss conditional on the portfolio having entered the tail beyond its VaR threshold.

Unlike VaR, which identifies a loss threshold, Expected Shortfall provides information about the magnitude of losses once that threshold has been exceeded.

Historical Expected Shortfall

At confidence level c , Historical Expected Shortfall is calculated as

$$ES_c = -E[R_p \mid R_p \leq Q_{1-c}(R_p)].$$

Thus, the 99% Expected Shortfall represents the average portfolio loss among observations lying in approximately the worst 1% of the historical return distribution.

Parametric Gaussian Expected Shortfall

Under the Gaussian assumption,

$$R_p \sim N(\mu_p, \sigma_p^2),$$

Expected Shortfall is

$$ES_c = -\mu_p + \sigma_p \frac{\phi(z_{1-c})}{1-c},$$

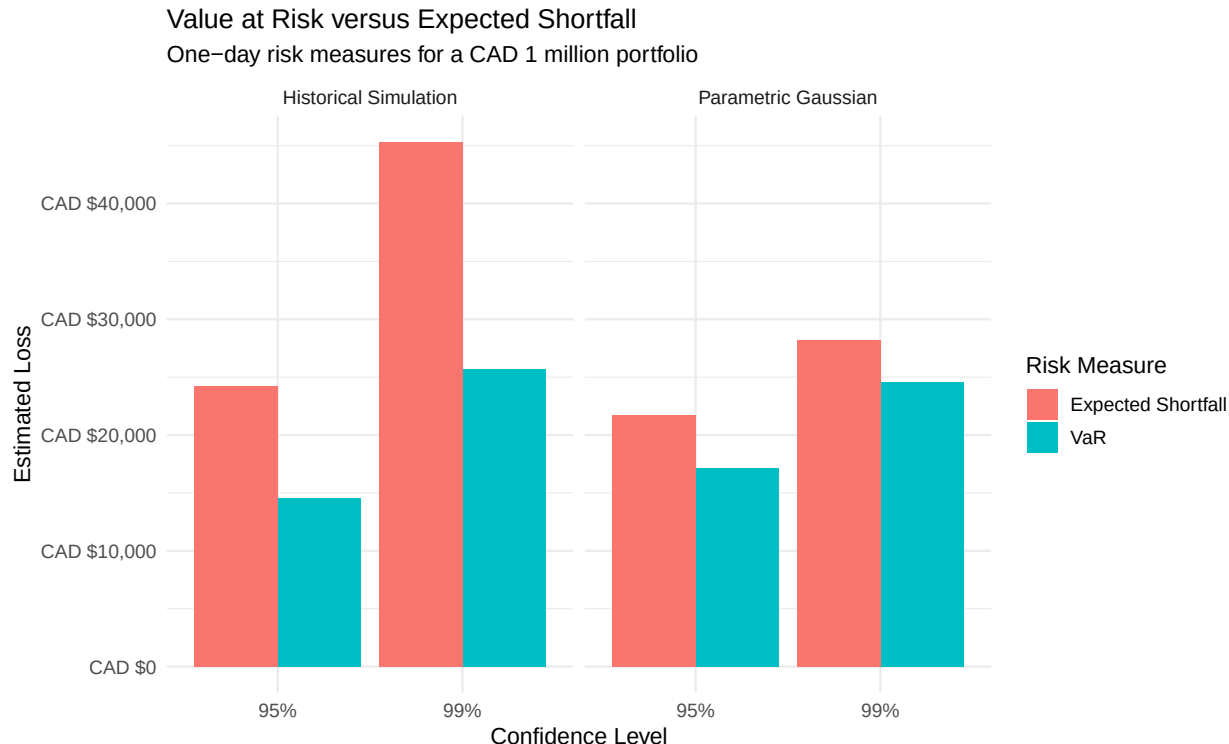
where $\phi(\cdot)$ is the standard normal probability density function.

Expected Shortfall Estimates

Table 8: One-Day Expected Shortfall Estimates

Model	Confidence Level	Expected Shortfall (%)	Expected Shortfall (CAD)
Historical Simulation	95%	2.42%	CAD \$24,196.51
Parametric Gaussian	95%	2.17%	CAD \$21,686.01
Historical Simulation	99%	4.53%	CAD \$45,335.28
Parametric Gaussian	99%	2.82%	CAD \$28,211.26

VaR and Expected Shortfall Comparison



Volatility Modelling

Daily financial-market volatility is unlikely to remain constant through time. The rolling-volatility analysis indicates periods of both elevated and subdued risk, motivating the use of conditional volatility models.

Two approaches are considered: Exponentially Weighted Moving Average (EWMA) volatility and a GARCH(1,1) model.

EWMA Volatility

The EWMA model updates conditional variance according to

$$\sigma_t^2 = \lambda \sigma_{t-1}^2 + (1 - \lambda) r_{t-1}^2.$$

The baseline model uses

$$\lambda = 0.94.$$

Recent observations therefore receive greater influence than distant observations.

GARCH(1,1)

The GARCH(1,1) conditional variance model is

$$\sigma_t^2 = \omega + \alpha \epsilon_{t-1}^2 + \beta \sigma_{t-1}^2.$$

The parameter α captures the response of volatility to new shocks, while β captures volatility persistence. The quantity

$$\alpha + \beta$$

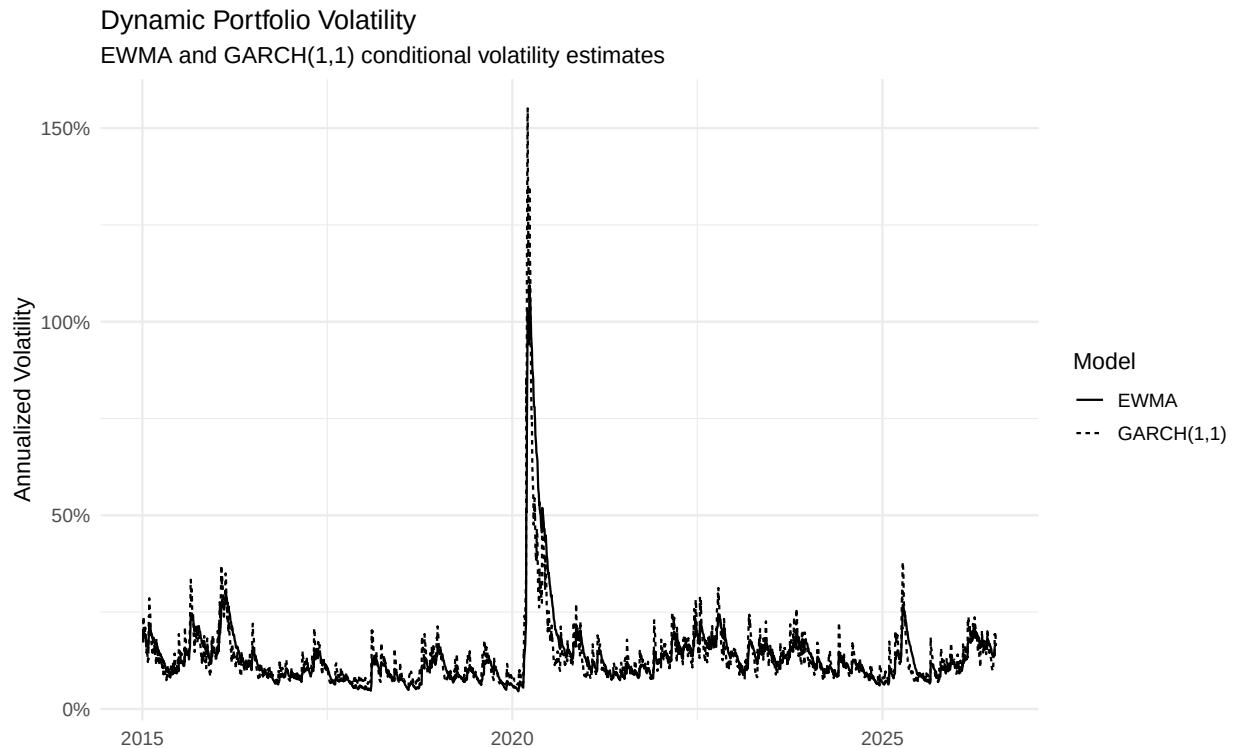
provides a useful measure of overall volatility persistence.

Estimated GARCH Parameters

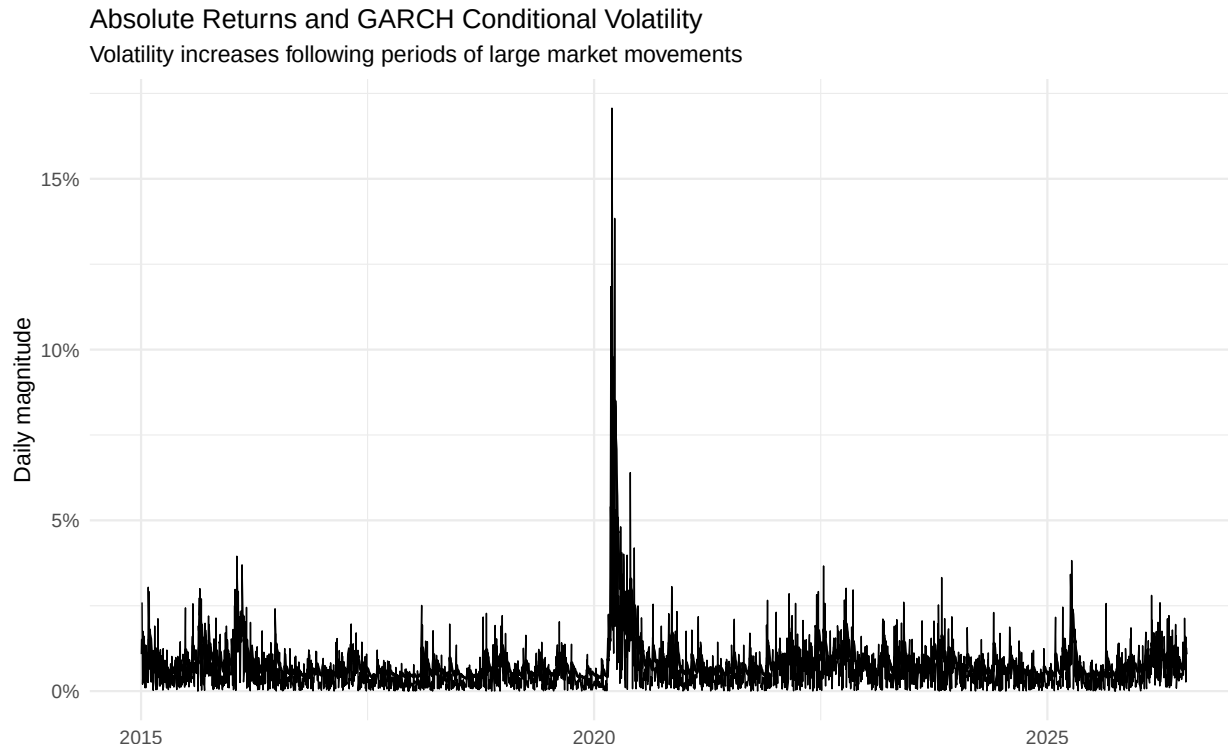
Table 9: Estimated GARCH(1,1) Parameters

Parameter	Estimate
mu	0.090755
omega	0.024501
alpha1	0.179149
beta1	0.799440

Conditional Volatility Through Time



Returns and Conditional Volatility



Backtesting

Out-of-Sample VaR Forecasting

The static VaR estimates presented earlier use the entire historical sample and are therefore primarily descriptive. To evaluate model performance more rigorously, one-day-ahead VaR forecasts are generated using only information available before each forecast date.

A rolling estimation window of 500 trading days is used.

For forecast date t , model parameters are estimated using returns from

$$t - 500$$

through

$$t - 1.$$

The resulting VaR forecast is then compared with the realized return on date t .

Four models are evaluated:

1. Historical Simulation
2. Rolling Gaussian VaR
3. EWMA Gaussian VaR

4. GARCH(1,1) Gaussian VaR

A VaR violation is recorded when the realized portfolio return falls below the model-implied loss threshold:

$$R_t < -VaR_t.$$

At the 95% confidence level, the theoretical violation probability is 5%, while at the 99% confidence level it is 1%.

Backtesting Summary

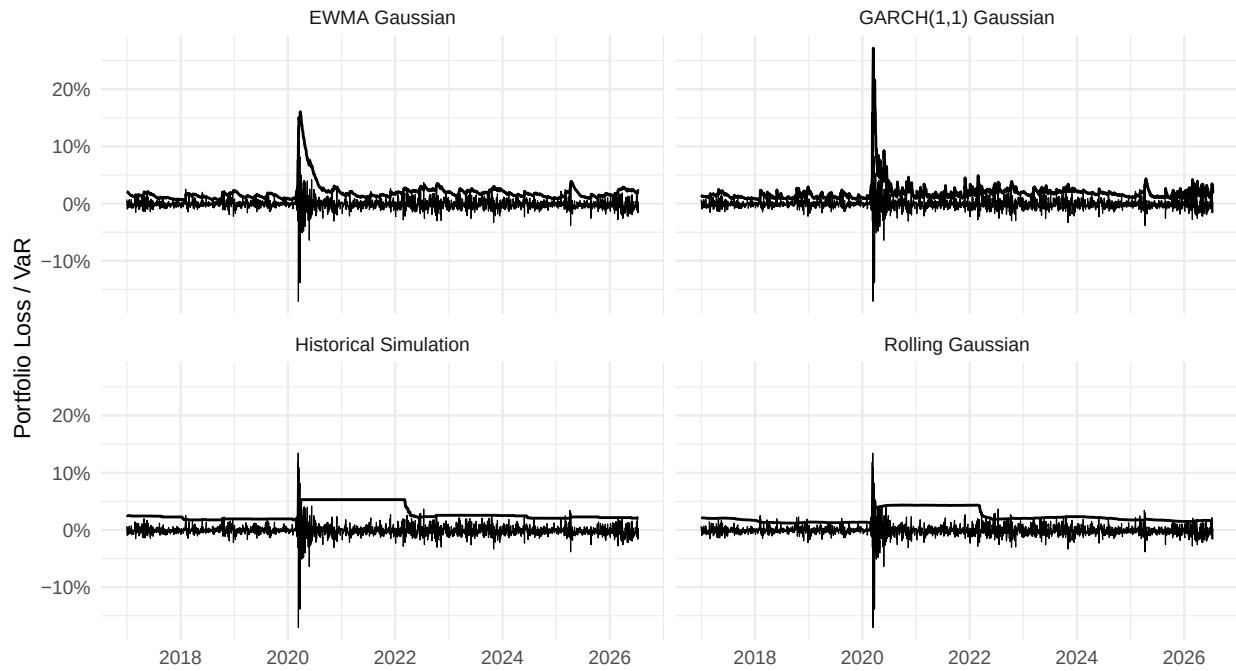
Table 10: Out-of-Sample VaR Backtesting Summary

Model	Confidence Level	Observations	Expected Violations	Actual Violations	Expected Violation Rate	Actual Violation Rate	Violation Ratio	Average VaR	Maximum VaR
EWMA	95%	2394	119.70	148	5.00%	6.18%	1.24	1.35%	11.36%
Gaussian									
GARCH(1,1)	95%	2394	119.70	159	5.00%	6.64%	1.33	1.30%	19.20%
Gaussian									
Historical	95%	2394	119.70	134	5.00%	5.60%	1.12	1.43%	2.10%
Simulation									
Rolling	95%	2394	119.70	129	5.00%	5.39%	1.08	1.60%	3.08%
Gaussian									
EWMA	99%	2394	23.94	71	1.00%	2.97%	2.97	1.94%	16.06%
Gaussian									
GARCH(1,1)	99%	2394	23.94	66	1.00%	2.76%	2.76	1.88%	27.17%
Gaussian									
Historical	99%	2394	23.94	28	1.00%	1.17%	1.17	2.85%	5.31%
Simulation									
Rolling	99%	2394	23.94	53	1.00%	2.21%	2.21	2.29%	4.36%
Gaussian									

99% VaR Through Time

One-Day 99% VaR Backtest

Realized portfolio losses versus model-implied VaR



99% VaR Violations

99% VaR Violations

Points identify realized returns below the forecast VaR threshold



Statistical VaR Validation

Violation counts alone do not provide sufficient evidence of VaR model adequacy. Formal likelihood-ratio tests are therefore applied to evaluate both the frequency and temporal dependence of VaR exceedances.

Three tests are considered:

1. Kupiec unconditional coverage;
2. Christoffersen independence;
3. Christoffersen conditional coverage.

Kupiec Unconditional Coverage

Let T denote the number of VaR forecasts, x the number of violations, and $p = 1 - c$ the theoretical violation probability.

The observed violation rate is

$$\hat{p} = \frac{x}{T}.$$

The Kupiec likelihood-ratio statistic tests

$$H_0 : \hat{p} = p.$$

Under the null hypothesis,

$$LR_{UC} \sim \chi_1^2.$$

Christoffersen Independence

The independence test evaluates whether VaR violations exhibit serial dependence. Let

$$I_t = \begin{cases} 1, & \text{if a VaR violation occurs,} \\ 0, & \text{otherwise.} \end{cases}$$

Transition counts $n_{00}, n_{01}, n_{10}, n_{11}$ are used to test whether the probability of a violation depends on the previous day's violation state.

Under the null hypothesis,

$$LR_{IND} \sim \chi_1^2.$$

Conditional Coverage

Conditional coverage jointly evaluates correct violation frequency and independence:

$$LR_{CC} = LR_{UC} + LR_{IND}.$$

Under the null hypothesis,

$$LR_{CC} \sim \chi_2^2.$$

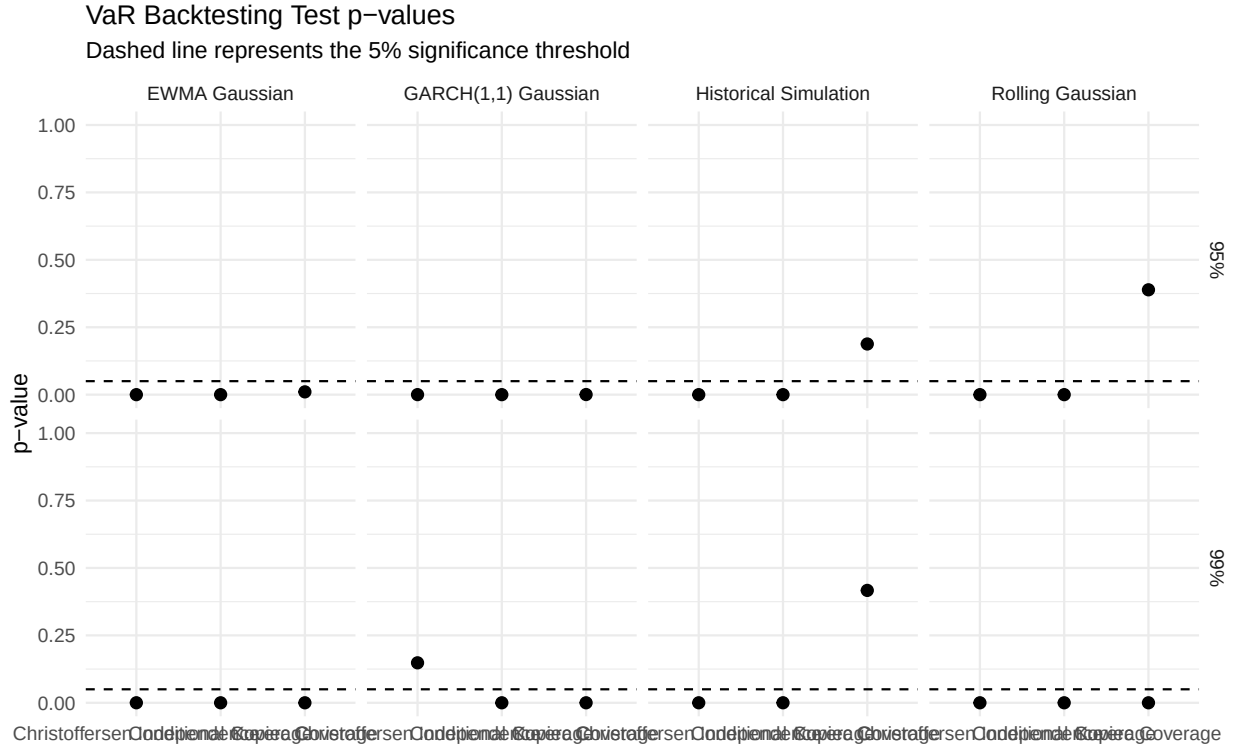
A 5% significance level is used throughout. Failure to reject a null hypothesis indicates that the available backtesting evidence is not sufficient to reject the corresponding model property.

Backtesting Test Results

Table 11: Formal VaR Backtesting Results

Model	Confidence Level	Expected Violations	Actual Violations	Kupiec p-value	Kupiec Re-sult	Independence p-value	Independence Result	Conditional Coverage p-value	Conditional Coverage Result
EWMA Gaussian	95%	119.70	148	0.010	Reject	0.000	Reject	0	Reject
GARCH(1,1) Gaussian	95%	119.70	159	0.000	Reject	0.000	Reject	0	Reject
Historical Simulation	95%	119.70	134	0.188	Do not reject	0.000	Reject	0	Reject
Rolling Gaussian	95%	119.70	129	0.389	Do not reject	0.000	Reject	0	Reject
EWMA Gaussian	99%	23.94	71	0.000	Reject	0.000	Reject	0	Reject
GARCH(1,1) Gaussian	99%	23.94	66	0.000	Reject	0.148	Do not reject	0	Reject
Historical Simulation	99%	23.94	28	0.417	Do not reject	0.000	Reject	0	Reject
Rolling Gaussian	99%	23.94	53	0.000	Reject	0.000	Reject	0	Reject

Statistical Backtesting Comparison



Stress Testing

VaR and Expected Shortfall characterize portfolio risk using an estimated return distribution. Stress testing provides a complementary perspective by evaluating portfolio performance under severe historical or hypothetical scenarios without requiring an explicit probability for each scenario.

Three forms of stress analysis are considered:

1. historical stress windows;
2. empirical extreme historical losses;
3. hypothetical security-level shocks.

Historical Stress Scenarios

Historical stress tests replay realized portfolio returns over predefined periods of elevated market stress.

For a scenario containing T daily returns, cumulative portfolio performance is calculated as

$$R_{\text{scenario}} = \prod_{t=1}^T (1 + R_{p,t}) - 1.$$

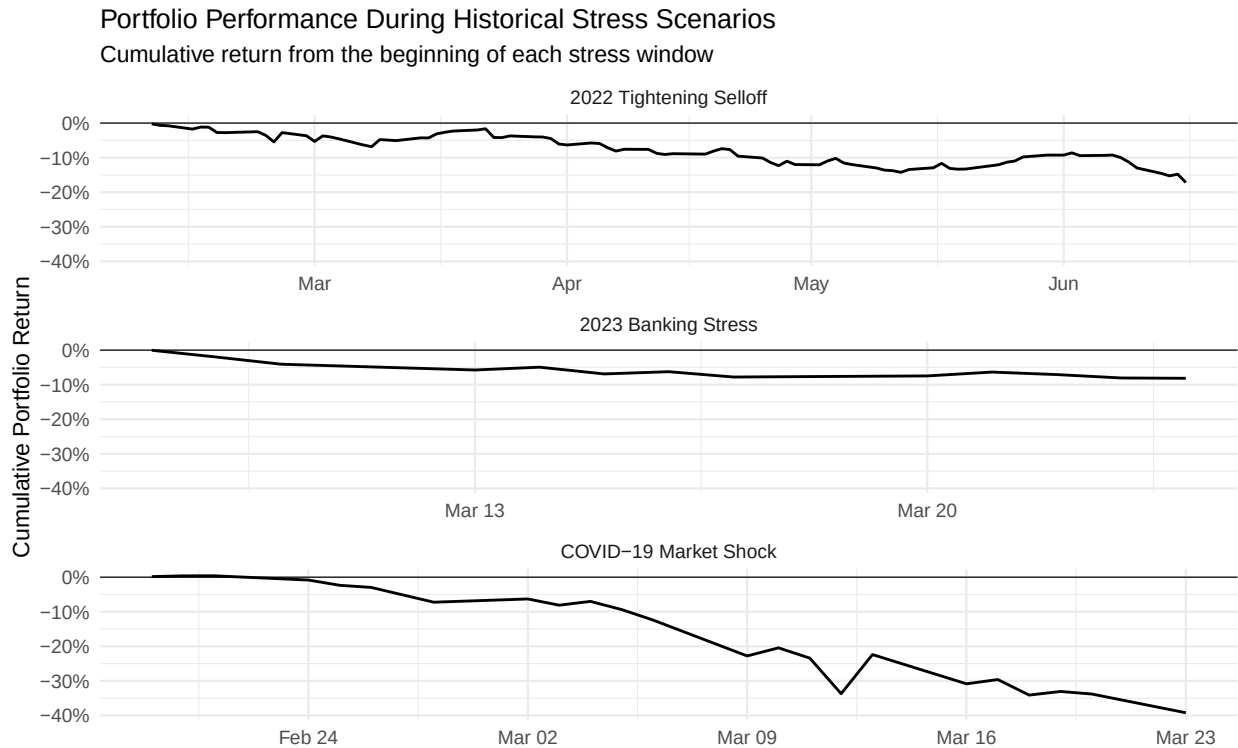
The corresponding portfolio P&L is

$$PnL_{\text{scenario}} = V_0 R_{\text{scenario}}.$$

Table 12: Historical Portfolio Stress Tests

Scenario	Start Date	End Date	Trading Days	Cumulative Return	Portfolio P&L	Worst Daily Return	Maximum Draw-down	Realized Volatility
COVID-19 Market Shock	2020-02-19	2020-03-23	24	-39.22%	-CAD \$392,163	-13.44%	-39.46%	94.89%
2022 Tightening Selloff	2022-02-09	2022-06-16	89	-17.18%	-CAD \$171,846	-2.82%	-17.18%	16.77%
2023 Banking Stress	2023-03-08	2023-03-24	13	-8.15%	-CAD \$81,543	-2.11%	-8.15%	19.11%

Portfolio Performance During Historical Stress



Empirical Extreme Losses

The historical sample is also searched mechanically for the worst one-day return and the worst five-trading-day compounded return.

Table 13: Most Severe Empirical Portfolio Losses

Scenario	End Date	Stress Return	Portfolio P&L	Stress Loss
Worst Historical 1-Day Loss	2020-03-12	-13.44%	-CAD \$134,355	CAD \$134,355
Worst Historical 5-Day Loss	2020-03-12	-26.84%	-CAD \$268,397	CAD \$268,397

Hypothetical Stress Scenarios

Hypothetical stress tests impose predetermined shocks directly on the six bank positions. For a vector of asset shocks $\mathbf{s}$,

$$R_p^{stress} = \mathbf{w}'\mathbf{s}.$$

The corresponding dollar impact is

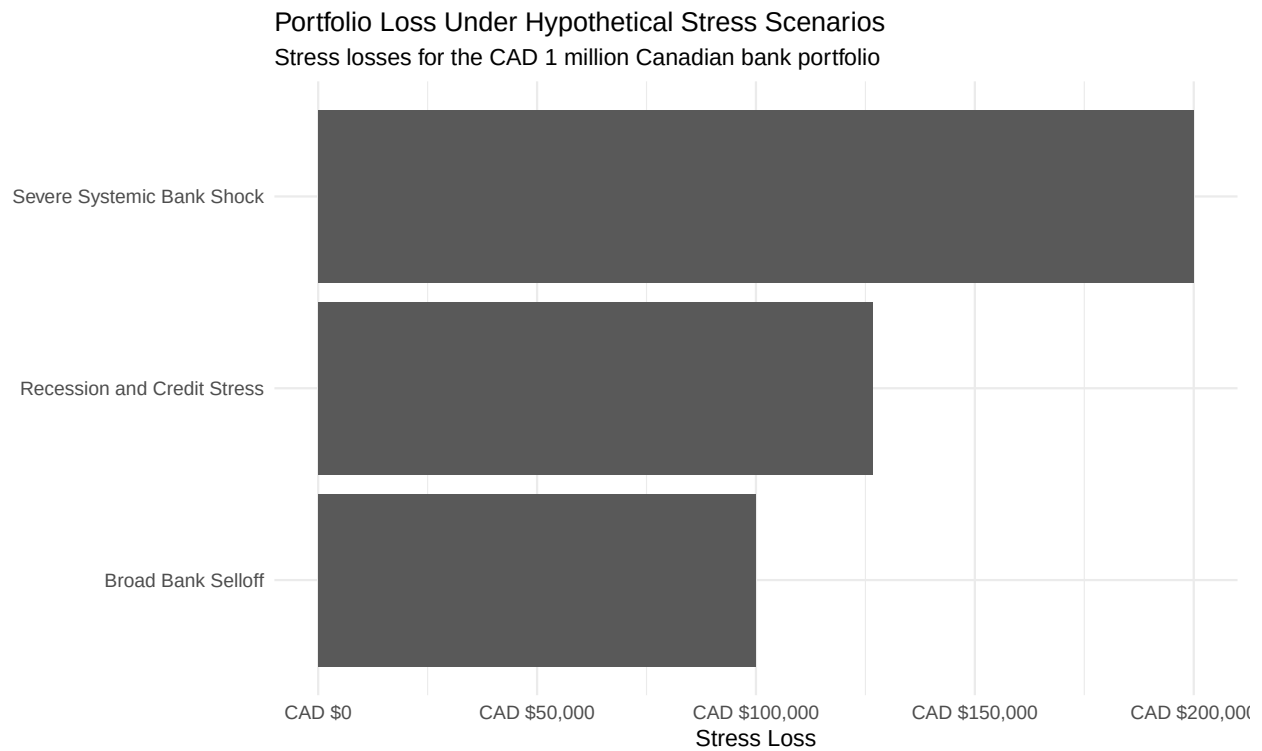
$$PnL_p^{stress} = V_0 R_p^{stress}.$$

The scenarios are deliberately severe and are intended for sensitivity analysis rather than as probabilistic forecasts.

Table 14: Hypothetical Portfolio Stress Tests

Scenario	Portfolio Stress Return	Portfolio P&L	Stress Loss
Broad Bank Selloff	-10.00%	-CAD \$100,000	CAD \$100,000
Recession and Credit Stress	-12.67%	-CAD \$126,667	CAD \$126,667
Severe Systemic Bank Shock	-20.00%	-CAD \$200,000	CAD \$200,000

Hypothetical Stress Loss Comparison



Risk Dashboard and Model Comparison

Latest Dynamic VaR Estimates

Table 15: Latest One-Day Dynamic VaR Estimates

Date	Model	Confidence Level	VaR (%)	VaR (CAD)
2026-07-15	EWMA Gaussian	95%	1.51%	CAD \$15,084.83
2026-07-15	GARCH(1,1) Gaussian	95%	1.30%	CAD \$12,951.80
2026-07-15	Historical Simulation	95%	1.19%	CAD \$11,917.75
2026-07-15	Rolling Gaussian	95%	1.14%	CAD \$11,397.53
2026-07-15	EWMA Gaussian	99%	2.20%	CAD \$22,041.80
2026-07-15	GARCH(1,1) Gaussian	99%	1.91%	CAD \$19,103.04
2026-07-15	Historical Simulation	99%	2.13%	CAD \$21,337.60
2026-07-15	Rolling Gaussian	99%	1.68%	CAD \$16,826.78

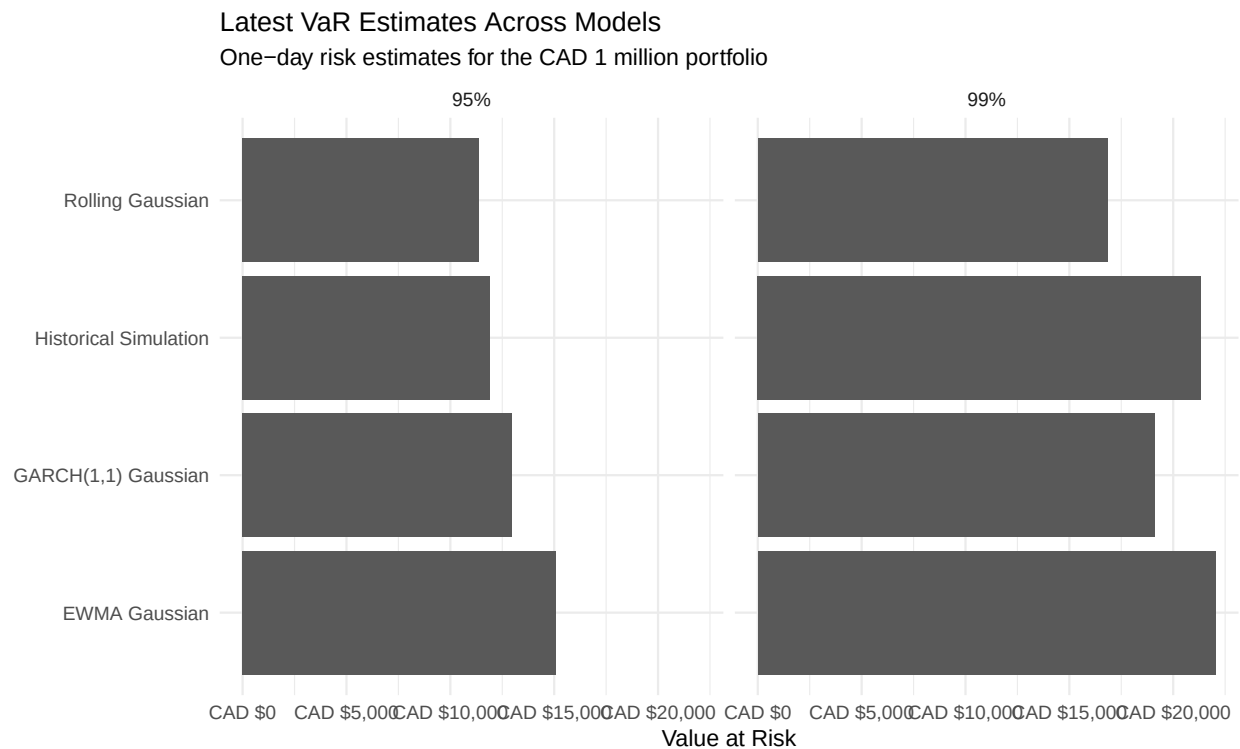
VaR Model Comparison

Table 16: VaR Model Performance and Validation Summary

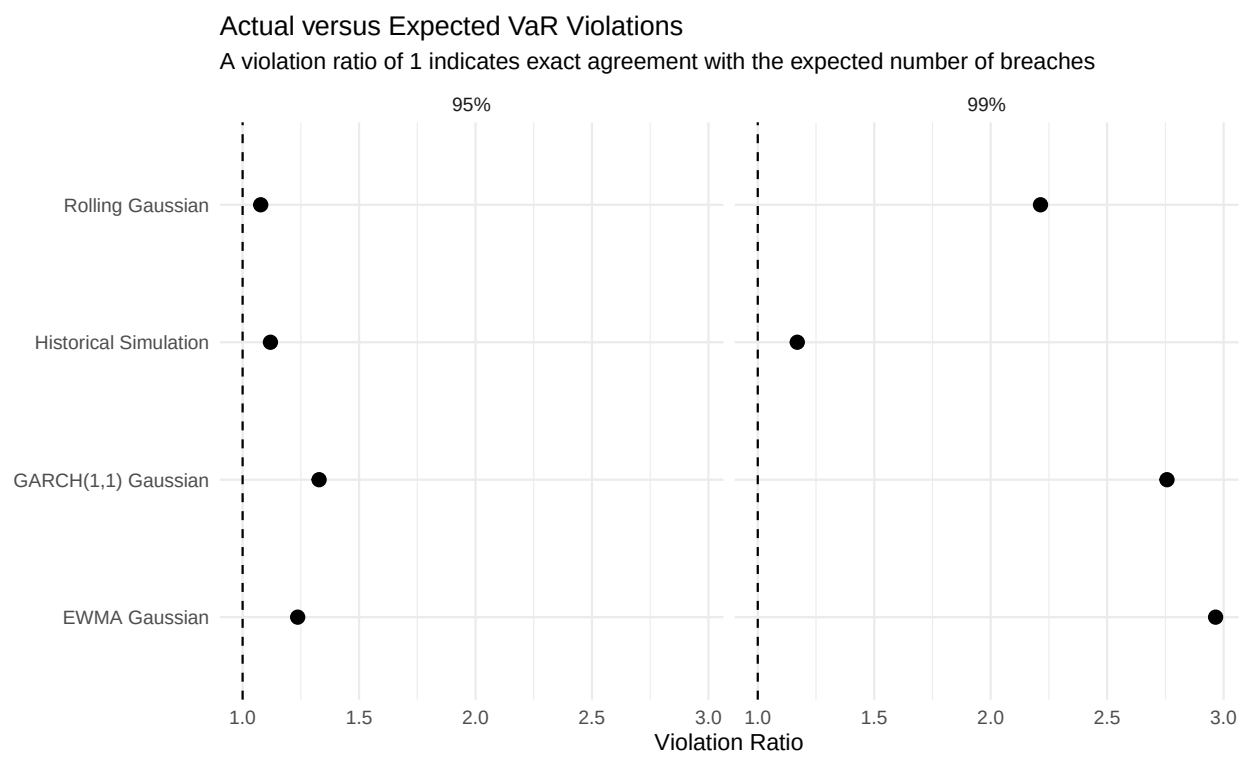
Model	Expected		Actual	Kupiec				Conditional		Tests Passed	Assessment
	Confidence Level	Violations	Violations	Violation Rate	Violation Ratio	Average VaR	p-value	Independence p-value	Coverage p-value		
Rolling Gaussian	95%	119.70	129	5.39%	1.08	1.60%	0.389	0.000	0	1	Mixed evidence
Historical Simulation	95%	119.70	134	5.60%	1.12	1.43%	0.188	0.000	0	1	Mixed evidence
EWMA Gaussian	95%	119.70	148	6.18%	1.24	1.35%	0.010	0.000	0	0	Weak statistical support
GARCH(1,1) Gaussian	95%	119.70	159	6.64%	1.33	1.30%	0.000	0.000	0	0	Weak statistical support
Historical Simulation	99%	23.94	28	1.17%	1.17	2.85%	0.417	0.000	0	1	Mixed evidence
GARCH(1,1) Gaussian	99%	23.94	66	2.76%	2.76	1.88%	0.000	0.148	0	1	Mixed evidence
Rolling Gaussian	99%	23.94	53	2.21%	2.21	2.29%	0.000	0.000	0	0	Weak statistical support

Model	Confidence Level	Expected Violations	Actual Violations	Violation Rate	Violation Ratio	Average VaR	Kupiec p-value	Independence p-value	Conditional Coverage p-value	Tests Passed	Assessment
EWMA Gaussian	99%	23.94	71	2.97%	2.97	1.94%	0.000	0.000	0	0	Weak statistical support

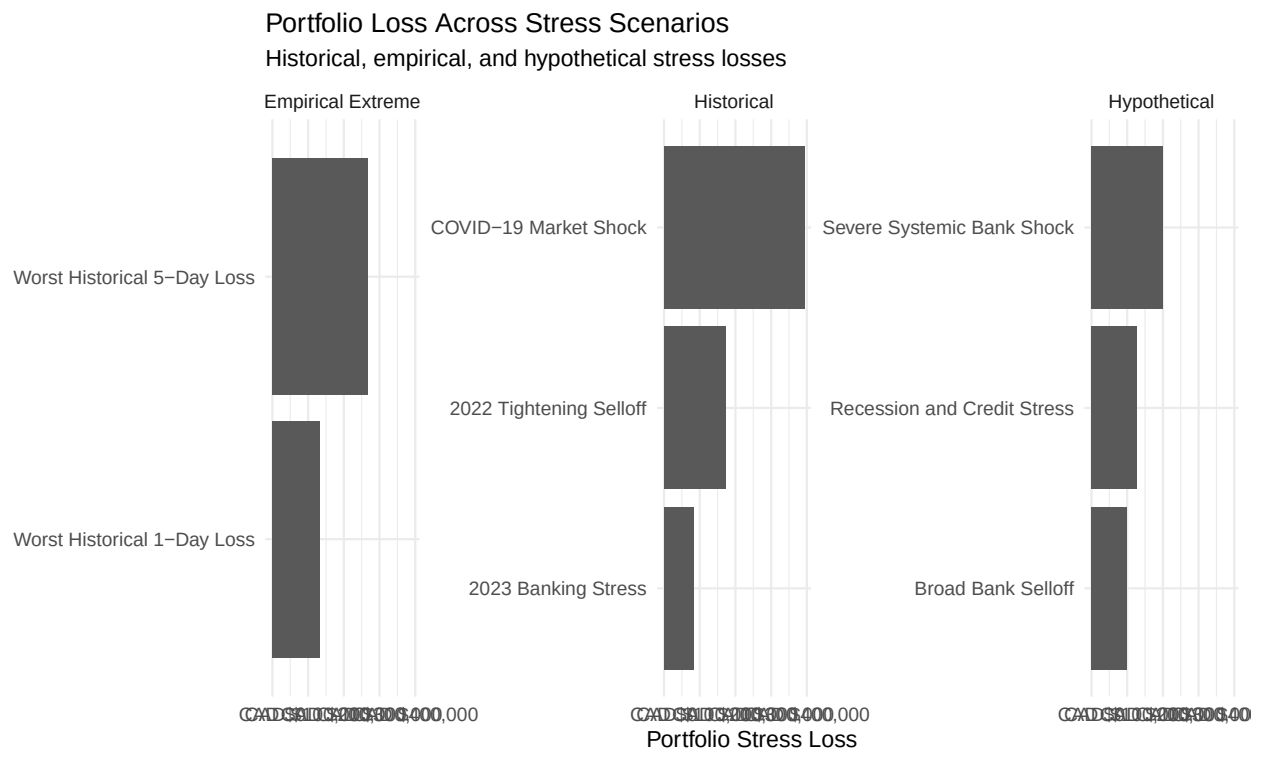
Latest VaR Across Models



VaR Calibration Across Models



Stress-Loss Comparison



Model Limitations

The results of the market risk engine should be interpreted subject to several important model and data limitations.

Distributional Assumptions

The Rolling Gaussian, EWMA, and baseline GARCH VaR specifications use Gaussian conditional return assumptions. Financial returns can exhibit negative skewness, excess kurtosis, and extreme observations that occur more frequently than implied by a normal distribution. Consequently, Gaussian models may underestimate tail risk even when their volatility forecasts are well calibrated.

Historical Simulation avoids an explicit parametric distributional assumption but is restricted by the empirical observations contained within the estimation window. It cannot naturally generate losses more severe than those already represented in the historical sample.

Parameter and Estimation Risk

VaR and volatility estimates depend on choices such as the estimation-window length, EWMA decay parameter, GARCH specification, innovation distribution, and model refitting frequency. Alternative specifications may produce materially different risk estimates.

The 500-trading-day rolling window represents a compromise between responsiveness and statistical precision. A shorter window may react more quickly to regime changes but produces noisier estimates, while a longer window may provide greater stability at the cost of slower adaptation.

Volatility Model Risk

EWMA assumes a fixed decay parameter of

$$\lambda = 0.94,$$

rather than estimating persistence directly from the portfolio data.

The GARCH(1,1) model provides greater flexibility by estimating volatility persistence, but its performance remains dependent on the selected variance equation and innovation distribution. More sophisticated specifications, including Student- t , asymmetric GARCH, or regime-switching models, could capture additional features of the return process.

Backtesting Limitations

Failure to reject a VaR model in the Kupiec or Christoffersen tests does not prove that the model is correct. Statistical tests have finite-sample limitations, particularly at high confidence levels where relatively few violations are expected.

For example, a 99% VaR backtest uses only approximately 1% of observations as expected violations. Consequently, inference concerning extreme-tail calibration is based on relatively limited information.

Backtesting results should therefore be interpreted as evidence regarding model adequacy rather than definitive proof of model validity.

Portfolio Concentration

The portfolio contains six individual securities but remains concentrated entirely within the Canadian banking industry.

Because bank returns can become highly correlated during systemic periods of stress, diversification across individual banks provides substantially less protection than diversification across independent sectors, geographies, and asset classes.

The engine therefore measures the risk of a deliberately concentrated Canadian-bank portfolio rather than a broadly diversified institutional portfolio.

Market and Liquidity Risk

The analysis is based primarily on closing-price returns and does not explicitly model:

- bid-ask spreads;
- market depth;
- transaction costs;
- price impact;
- liquidity deterioration;
- trading halts;
- execution delays.

During stressed markets, these effects can increase realized losses beyond those implied by closing-price VaR calculations.

Stress-Scenario Limitations

Historical stress tests are constrained by events that occurred during the available sample period, while hypothetical scenarios depend on analyst judgement.

Stress scenarios should therefore be interpreted as sensitivity analyses rather than forecasts with assigned probabilities.

Model Scope

The engine focuses on equity market risk. It does not separately model the banks' underlying exposures to credit risk, interest-rate risk, funding risk, foreign-exchange risk, operational risk, or liquidity risk.

The results should therefore not be interpreted as measures of the total enterprise risk of the underlying financial institutions.

Risk Management Recommendations

Based on the combined VaR, Expected Shortfall, volatility, backtesting, and stress-testing framework, the following risk-management practices are appropriate for the portfolio.

1. Use Dynamic VaR for Day-to-Day Monitoring

Conditional volatility models such as EWMA and GARCH should be considered alongside static risk measures because portfolio volatility changes through time. Dynamic models can respond more rapidly when market conditions deteriorate.

2. Maintain a Challenger-Model Framework

No single VaR methodology should be treated as uniquely correct. Historical Simulation, Gaussian, EWMA, and GARCH estimates should be compared regularly, particularly when their risk estimates diverge materially.

Large differences between models can themselves provide useful information about model uncertainty.

3. Monitor Expected Shortfall Alongside VaR

VaR identifies a tail threshold but provides no information regarding the severity of losses beyond that threshold. Expected Shortfall should therefore be monitored as a complementary tail-risk measure.

4. Treat Backtesting as an Ongoing Validation Process

Violation frequencies and clustering should be reassessed whenever new observations become available. Persistent rejection of unconditional or conditional coverage would indicate the need to recalibrate or reconsider the affected model.

5. Maintain Stress Limits Separately from VaR Limits

Severe historical and hypothetical losses can be substantially larger than model-implied daily VaR. Stress-testing limits should therefore complement rather than replace ordinary VaR monitoring.

6. Recognize Sector Concentration Explicitly

The portfolio's six positions are all exposed to common Canadian banking-sector risk factors. A broader portfolio would require diversification across sectors, geographies, and potentially asset classes to reduce systemic concentration.

7. Reassess Models During Structural Change

Risk-model parameters estimated from historical data may become less representative following changes in monetary policy, regulation, financial-market structure, or macroeconomic conditions. Model performance should therefore be reviewed when significant structural changes occur.

Conclusion

The Canadian Bank Market Risk VaR Engine demonstrates how multiple quantitative risk methodologies can be integrated into a single reproducible framework for portfolio market-risk measurement and model validation.

The analysis begins with adjusted market prices and constructs a constant-weight portfolio return series for six major Canadian banks. Static Historical Simulation and Gaussian VaR provide baseline measures of downside risk, while Expected Shortfall extends the analysis by quantifying the severity of losses beyond the VaR threshold.

The empirical return analysis demonstrates that market risk is dynamic rather than constant through time. EWMA and GARCH(1,1) models address this feature by producing conditional volatility estimates that respond to changing market conditions.

The rolling backtesting framework represents a particularly important improvement over full-sample risk estimation. By generating each VaR forecast using only information available before the realized return, the analysis permits genuine out-of-sample evaluation. Kupiec and Christoffersen tests then evaluate whether VaR violations occur with the appropriate frequency and whether those violations exhibit statistically significant clustering.

At the 99% confidence level, **Historical Simulation** receives the highest descriptive ranking in the current validation framework, passing 1 of three statistical backtesting criteria and producing a violation ratio of approximately 1.17. This result represents evidence in favor of the model within the present sample and specification, rather than proof that it will remain superior under future market regimes.

Stress testing demonstrates why VaR should not be used as a complete measure of portfolio downside risk. The largest stress loss evaluated by the engine is approximately **CAD \$392,163**, which can materially exceed normal one-day VaR estimates. Expected Shortfall and stress testing therefore provide essential complementary information concerning the severity of extreme losses.

From a risk-management perspective, the principal conclusion is that no single model should be relied upon in isolation. Dynamic VaR provides useful short-horizon monitoring, Historical Simulation supplies a non-parametric challenger model, Expected Shortfall measures extreme-tail severity, and stress testing evaluates losses outside the assumptions of the probabilistic VaR framework.

The portfolio also remains materially exposed to sector concentration. Holding six Canadian banks provides diversification across issuers but limited protection against systemic shocks affecting the Canadian financial sector as a whole.

Overall, the engine illustrates a model-risk principle that is central to market risk management: risk estimates are conditional on the data, assumptions, and methodology used to produce them. Effective risk management therefore requires not only calculating VaR, but validating it, challenging it with alternative models, examining losses beyond VaR, and evaluating portfolio resilience under severe stress.